(a) input page

the ViT encoder and LLM decoder through parallel parameter reuse. ParVL constructs separate prefix-conditioned branches for the two components, aggregates their representations at modality-specific stages, and controls their branch counts independently. This design enables expanded computation to be allocated flexibly between vision and language with modest parameter growth.

Under the same full-parameter SFT recipe, the best evaluated ParVL configuration improves the 9-benchmark average from 49.6 to 50.5 at 1B. At the 2B and 8B scales, the balanced branch-count configuration ($P_v = 2$, $P_l = 2$) yields average scores of 54.7 and 63.0, compared with 54.4 and 62.5 for the corresponding single-branch baselines.

Our contributions are threefold. (1) We introduce ParVL, a parallel-scaling framework for MLLMs that formulates computation expansion as a two-dimensional vision–language allocation problem and implements it with parameter-shared, prefix-conditioned ViT and LLM branches and modality-specific token-wise aggregation. (2) Through a systematic study of nine vision–language branch-count configurations, we show that the preferred compute allocation varies across benchmarks. (3) Controlled comparisons at 1B, 2B, and 8B scales, together with latency and memory profiling, characterize the accuracy–cost trade-offs of parallel scaling.

Related Work

Scaling Laws for Language Models. Scaling laws characterize predictable relationships between model performance and parameter count, training-data size, and training compute (Kaplan et al. 2020; Hoffmann et al. 2022). Dense parameter scaling increases capacity but also raises storage and memory costs. Sparse MoE models instead expand total

new scientific question that does not exist in the text-only domain: given a fixed parameter budget, how can one effectively expand the model's computational capacity via parameter reuse, and concurrently determine how computational resources should be allocated between the vision (ViT) and language (LLM) modalities? In this work, we are the first to systematically investigate this vision-language computation allocation trade-off. We adapt a parallel architecture inspired by ParScale merely as a *vehicle* for this study, proposing our ParVL framework to flexibly control and evaluate this novel resource allocation problem.

Inference-Time Scaling. The success of reasoning-oriented language models, exemplified by DeepSeek-R1 (Guo et al. 2025), has intensified interest in inference-time scaling, which allocates additional computation during inference to improve model performance. Serial inference-time scaling commonly allocates additional computation through longer Chain-of-Thought reasoning traces (Wei et al. 2022), thereby extending the serial inference path and typically increasing latency. Instead, parallel scaling expands the inference computation across multiple candidate paths. The beam search maintains multiple partial hypotheses during decoding (Wiseman and Rush 2016), whereas Self-Consistency samples diverse Chain-of-Thought trajectories and aggregates their final answers (Wang et al. 2022). Where hardware resources permit, candidate paths can be evaluated concurrently, so increasing inference computation does not need to translate linearly into wall-clock latency.

However, concurrency does not eliminate redundancy across independently generated paths, motivating methods that coordinate parallel reasoning or prune redundant trajectories. Adaptive parallel reasoning learns to invoke

(b) fused blocks (2 seen by both detectors)

(c) reading order + 4 chunks